

Forecasting of the ID_3 using limit order book data

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Forecasting of electricity intraday market indices has become an actively researched field in recent years. Nonetheless, none of the existing approaches incorporate information that is finer than the transaction data that the indices are based on. We analyse whether the usage of data derived from the limit order books of the underlying products can enhance the index forecasts. In order to do so, we first specify possible order book characteristics to use for forecasting such as mid prices and bid-ask spreads and select from them using elastic nets. A naive model as well as a simple regression model are created based on the results. Our findings suggest that time-weighted mid prices do contain relevant information for forecasting indices, which holds true even more for quarter hourly than for hourly products.

1. Introduction

In 2010, the German government issued plans regarding the electricity supply in Germany for the coming decades [19]. Its goals were to decrease greenhouse gas emissions as well as dependencies from electricity imports while keeping electricity supply stable, sustainable and affordable. These goals all have one solution component in common: An increase of renewable electricity production locally in Germany. The governmental plan thus included to enlarge the share of renewable electricity in respect of gross electricity consumption to 80% in 2050. The release of this goal led to a surge in the expansion of renewable electricity generation capacity in Germany over the last decade. As a consequence, the share of renewable electricity in respect of gross electricity consumption increased from 17% in 2010 to an astonishing 45.4% in 2020 already. This increase has indeed been stronger than before, as in the decade from 2000 to 2010, the share only rose from 6.7% to 17% [17].

The main sources for this renewable electricity rise in Germany are wind and solar energy [17]. From all possible sources of renewable electricity, these two are the ones with the most volatile infeed pattern. Solar electricity heavily depends on whether clouds or other shadows reduce the amount of possible solar electricity produced. Wind on the other hand is hard to predict, as every wind farm location and every wind turbine height has its own physical circumstances. The

fact that these two sources are the main drivers for renewable electricity production in Germany thus leads to new challenges for the coupling of electricity production and demand. Demand is usually quite dull and does not immediately react to price signals on the market from e.g. a forecast change for renewable infeed. The Day-Ahead market provided by EPEX SPOT, an auction where traders can buy or sell electricity for the next day, is not close enough to the start of delivery to react to these changes. This is where the electricity intraday market comes into play: It can be used to run and plan power generation close to delivery and therefore close to a better forecast of consumption as well as to adjust for forecast errors or unforeseen events [16]. These balancing features in combination with the increase in renewable production led to the electricity intraday market mirroring the development of renewable electricity production and continuously growing over the last years. In the year 2020 alone, the continuous intraday market operated by EPEX SPOT for the market area Germany/Luxembourg grew by 19.6% [11]. This indicates that a functioning and flexible electricity intraday market is a necessity for a functioning growth in volatile renewable electricity production.

In order to keep up with the increasing need to trade shortly before delivery, the EPEX SPOT electricity intraday market has continued to develop over the last years and EPEX SPOT expanded its service portfolio [13]. Traders can nowadays buy or sell hourly, half hourly or quarter hourly products on the market. The market principle is pay-as-bid, i.e. orders are collected and matched when possible during trading hours. Order collection and matching is managed in so-called *limit order books*. This market structure leads to many different occurring prices for the same product, which complicates the assessment of the product's "actual" price. In order to facilitate an evaluation of this actual price, several price signals in form of indices have been introduced for the traders on the intraday market [14]. All these indices focus on materialized trades for the different products on the market. Two of these indices take into account price evolutions shortly before delivery e.g. due to forecast corrections: the ID₃ and the ID₁. They are limited to the last three hours and last hour of national trading, respectively. As the ID₃ is the index that is used as underlying for German Intraday Cap and Floor Futures and most of the existing literature analyses it, we will concentrate on it in this paper.

Prediction of electricity intraday and especially index prices in Germany is a relatively new research topic. Nonetheless, a few recent papers address forecasting of intraday prices. [6] focus on a small renewable energy producer that wants to place an optimal amount of her electricity on the Day-Ahead and the rest on the intraday market. Thus, the spread between Day-Ahead prices and the weighted average of all intraday trades for each hourly contract is forecasted. [18] present a thorough study of intraday electricity forecasting with a focus on ID₃-prices. They apply the least absolute shrinkage and selection operator (LASSO) to an abundance of variables and compare ten of the emerging models with two benchmarks. A major takeaway is that the most recent intraday price as well as the Day-Ahead price for the specific hour seem to have the highest prediction power, with many other variables by far not as relevant. [9] follow a similar path and compare different full-information econometric time series models for all ID₃-prices, that are also reduced through the usage of LASSO and elastic nets. All resulting models are used for forecasting and compared with each other as well as benchmark models in terms of their forecast performance. They find that for hourly products, surprisingly none of the full-information models outperforms one of their simple naive benchmarks. [4] concentrate on forecasting the whole price distribution for the ID₃ instead of only the expected value. Their models manage to outperform the naive benchmarks in the tails of the distribution, but also show no significantly better results for the expected value. Finally, [7] use principal component analysis to average over a pool of forecasts for ID₃-prices to enhance the overall forecasting ability. Their model of the ID₃ relies on the usage of lagged ID₃-prices as well as Day-Ahead prices, if no lagged prices

are available. Their findings suggest that no forecast averaging outperforms the forecast based on the longest calibration window.

Our approach to electricity index price forecasting relies on the fact that the transaction data tracked by the indices are always a result of the underlying limit order book movements. We examine how data derived from the limit order book can be used to enhance these forecasts. Our findings indicate that lagged time-weighted mid prices, i.e. time-weighted mean prices between best ask and best bid prices from a time interval before the index interval, are a valuable source of information for forecasting ID₃ hourly and quarter hourly prices, with a tendency to contain even more information about the index prices than lagged volume-weighted transaction prices themselves. As the topic of forecasting requires usage of information that is adapted to the time of the forecast, we will omit the affix “lagged” from here on.

The rest of the paper is structured as follows: In chapter 2, we shortly introduce the German electricity markets as well as how the ID₃ is defined on the electricity intraday market. Chapter 3 gives insight into limit order book data and statistics that can be derived from it. Furthermore, we present an algorithm that is used to extract that data from the limit order book. This is followed by a presentation of the used data set and our ID₃-price model in Chapter 4, which also contains a real-world comparison with another model. Finally, chapter 5 concludes the paper by summing up the results and presenting further ideas.

2. Electricity intraday market and indices

German electricity trading takes place on three separate markets that we introduce here shortly, with their order being determined by how far away the delivery start of a product is during trading: The power derivatives market consists of trading of futures and options, that are concerned with products with delivery further in the future. This is complemented by a day-ahead market consisting of an auction with gate closure at 12 pm where hourly delivery contracts for the next day are traded. Finally, the continuous electricity intraday market opens at 3 pm and operates closest to the actual delivery, with products being traded up to five minutes before delivery. We will look at the intraday market operated by EPEX SPOT in the following. Positions that are still unbalanced at time of delivery are covered on the balancing market.

The intraday market The intraday market itself also offers an intraday auction for quarter hourly products, but the part that we will concentrate on in this paper is the continuous trading of products. There, market participants can buy or sell electricity contracts with quarter hourly, half hourly or hourly delivery. Continuous trading of hourly products of the next day begins at 3 pm and is soon followed by the trading start of half hourly (3:30 pm) and quarter hourly (4 pm) products. Cross-border trading through the usage of Single Intraday Coupling (SIDC), which allows trading with neighbouring countries in Europe, starts at 6 pm and ends one hour before delivery [15]. Every product can furthermore be traded nationally up to 30 minutes before its delivery, and in the same transmission system operator zone even up to five minutes before delivery. This trading stopping time is also called lead time in the literature. As mentioned before, the continuous intraday market works as a pay-as-bid market, where orders are collected in so-called limit order books, that are detailed in Section 3, and matched if possible. Therefore, the price paid for a specific product heavily depends on the momentary supply and demand for it as well as on market liquidity.

Electricity indices To give traders *signals* about the actual price of the products during different trading periods, several volume-weighted transaction price indices exist, each limited to a different time interval. The most popular one is called ID_3 and is used as underlying for German Intraday Cap and Floor Futures, see e.g. [3]. It is defined as volume-weighted average transaction price of all transactions that appeared during the last three hours of national trading, i.e.

$$ID_3^{d,i} = \sum_{t \in T^{d,i}} \frac{p_t v_t}{\sum_{t \in T^{d,i}} v_t}. \quad (1)$$

Here, d specifies the specific timestamp including date and hour of the delivery start, and $i \in \{h, qh\}$ denotes the type of the product, i.e. whether it is an hourly or quarter hourly product. The ID_3 also exists for half hourly products, but as we will not consider them in this paper, they are left out intentionally. Furthermore, $T^{d,i}$ contains all transactions of the product with delivery start at d and type i that appeared during the ID_3 period. As this period depends on the national and SIDC trading only, it is e.g. for the ID_3 of the hourly product delivering from 8 am to 9 am given by the time interval 4:30 am - 7:30 am. Finally, p_t is the price of transaction t and v_t is the corresponding traded volume. As the ID_3 depends on the prices and volumes that were matched on the market in the most active trading period of each product, it can be considered as a volume-weighted mean of prices that buyers are willing to pay and sellers are willing to take close to delivery.

3. Limit Order Book - data and derived data

During the trading window of every product, orders for it can be placed on the market. The trader chooses between placing a *limit order* or a *market sweep order*, in the literature often called market order, see [12].

A limit order contains a volume and a limit price at which the trader would buy or sell the product, with the option to get a better price. If the incoming limit order does not match an already existing order on the other side, it is placed into the limit order book (LOB). The LOB contains all orders that are neither immediately (and fully) matched nor canceled. When a new limit order matches one (or several) orders inside the book, the latter is (are) removed from the LOB and the initiating one does not enter the LOB at all. It is also possible that an order is only partly matched. Then, the remaining volume is again entered into the LOB.

In contrast to a limit order, a market sweep order contains only a volume and will always be filled completely except for the case that not enough volume is contained in the whole LOB. The price of the market order is composed by the volume-weighted prices from the limit orders that were matched. As a market sweep order is always filled as much as possible and unexecuted quantity is canceled, it never enters the LOB [12]. In everyday business, market orders tend to be replaced by limit orders with their price equaling the best price of the opposing market side. In conclusion, only those orders enter the LOB that actually provide liquidity to the market; those, that instead use the provided liquidity do not enter the book. More about the exact structure of an order that is entered into the market can be found in Appendix A.

Each product has its own LOB where orders for it are collected, and ordering as well as matching in the LOB is based on a price-time ranking: a limit order is sorted into the LOB first by checking its price, and, if another order with the same price is already part of the LOB, its execution priority is ranked lower than that of the existing order. Orders with the same price form a *price level*, and the *LOB depth* is defined as the number of different price levels. A match of the orders contained in the book with a new incoming order will always happen first for those orders with the best prices. Consequently, the buy order with the highest buy price is called *best bid* and defined as bid_t^* for time t , and the sell order with the lowest sell price is called *best ask* and

respectively defined as ask_t^* for time t . These two orders determine the prices a new bidder has to outbid in order to be placed first in the book, and also determine the price range that needs to be crossed by a new order in order to match. The arithmetic mean m_t between best bid and best ask is called *mid price* m_t , and the difference between them is called *bid-ask spread* s_t , see e.g. [5]. Therefore, we have

$$m_t := \frac{\text{ask}_t^* + \text{bid}_t^*}{2}, \quad (2)$$

$$s_t := \text{ask}_t^* - \text{bid}_t^*. \quad (3)$$

The concept of looking at mid prices and bid-ask spreads is known from stock price analysis: The mid price reflects the current value of the product, while the bid-ask spread yields an indicator for how much liquidity is on the market as well as how high the incurred transaction costs are. Transaction costs in this sense arise from the fact that an order in the book provides liquidity, and an order matching it demands and consequently removes liquidity from the market, which is paid through the bid-ask spread as well as through the difference between the best and the realized price.

All the data that is gathered during this process from the LOBs is commercially available from EPEX SPOT. As anonymity of the trading parties is guaranteed from the exchange, this data does only contains order information from which no inferences about the trading parties can be drawn. For a detailed introduction to the LOB data format, the reader is again referred to Appendix A. Roughly speaking, an order contains identification numbers, several timestamps, prices and volumes. These can be used to recreate the different states that the LOB attains throughout its life time. In Algorithm 1, we do exactly that and present how to recreate the LOB for a certain product.

Moving step by step through Algorithm 1, first, all orders that are entered into the trading system on the exchange are collected in *ob_data*. Before processing these trades, they are first filtered (line 4) such that the result only contains orders regarding the prespecified product. All orders that are specified as block orders for that product are removed from the data, as they are traded and matched in a different book. Then, all orders with a live span of zero milliseconds that are not matched are removed (line 5), as they do not enter the LOB. This concludes the preselection of the data to be processed. For every remaining order, two events are constructed: its activation event and its deactivation event, both with the corresponding timestamp of beginning and end of the order's lifetime, respectively (line 6). All these events are collected in *all_events*. After sorting *all_events* to let them appear in correct order of appearance, the events are processed one by one. It is checked whether they enter the LOB (which is the case if they do not match with an existing order in the book) or match and therefore remove an order from the order book. The detailed procedure is illustrated from the perspective of a buy order: The first check verifies if the corresponding event is an activation (line 10). If the answer is affirmative, the second check verifies whether the offered price is equal to or higher than the momentary best ask, in combination with a check whether the order is marked as executed (line 11). This indicates that the order is a market order, and these do not enter the LOB. Consequently, the corresponding deactivation event is marked such that it is omitted later on (line 13). If one of the conditions in the second check were not fulfilled, the order is entered into the LOB by following the price-time ranking (line 17). Finally, in the case where the first check was not affirmative, the event is a deactivation of an order, wherefore the corresponding element in the LOB is deleted (line 19). This procedure leads to full information about all states of the LOB at every point in time during its lifetime.

Algorithm 1: Order book building

Result: The output is a collection of all order book states for a certain delivery product over its lifetime.

```
1 begin
2   Initialize the lists buy_OB, sell_OB and the numerics best_ask, best_bid.
3   Read order book data as ob_data.
4   Filter ob_data such that only rows with fitting delivery product remain.
5   Rows with ob_data["validity_duration"] = 0 & ob_data["is_executed"] = 0 are
      removed.
6   Split each row in ob_data in two events (activation of the order and deactivation of
      the order). Define all_events as the data frame with all those events.
7   Sort all_events by start_validity_time, order_id, activation.
8   for row in all_events do
9     if row["side"] = "buy" then
10      if row["activation"] = 1, i.e. is an activation event then
11        if row["price"] ≥ best_ask & row["is_executed"] = 1 then
12          row contains a market order and is not entered in the order book.
13          The corresponding deactivation event with the same order_id is
            marked s.th. it is omitted later on.
14        else
15          buy_OB gains a new list entry with the name being the timestamp
            row["start_validity_time"] and the content being the list entry before
            this one plus row.
16      else
17        buy_OB gains a new list entry with the name being the timestamp
            row["start_validity_time"] and the content being the list entry before
            this one minus the entry belonging to row.
18      Update best_buy if necessary.
19    else
20      if row["activation"] = 1, i.e. is an activation event then
21        if row["price"] ≤ best_bid & row["is_executed"] = 1 then
22          row contains a market order and is not entered in the order book.
23          The corresponding deactivation event with the same order_id is
            marked s.th. it is omitted later on.
24        else
25          sell_OB gains a new list entry with the name being the timestamp
            row["start_validity_time"] and the content being the list entry before
            this one plus row.
26      else
27        sell_OB gains a new list entry with the name being the timestamp
            row["start_validity_time"] and the content being the list entry before
            this one minus the entry belonging to row.
28      Update best_ask if necessary.
29    end for
30    return buy_ob, sell_ob.
31 end
```

4. Forecast

In the introduction, we quoted several papers dealing with ID₃-price forecasting. The best performing models use a huge load of explanatory variables, amounting from up to 372 in [18] to up to 26259 in [9]. Model reduction and selection is managed through the usage of the least absolute selection and shrinkage operator (LASSO) method or through elastic nets. Neither in these huge sets of explanatory variables nor in any of the other papers though, other LOB information aside from transaction data is used. Therefore, we want to investigate whether some of the LOB statistics introduced in section 3 could actually enhance ID₃ forecasting performance. In order to answer that question, we first introduce interval versions of the LOB statistics described above for further usage.

Time-weighted LOB statistics From the LOB recreation that is available after the application of Algorithm 1, several derived data like mid prices, bid-ask spreads and order book depths can be calculated for every snapshot during the LOB's life span. Usually, a collection of these values for every change in the LOB amounts to a very high number of entries, most being valid for only seconds. Therefore, we aggregate them over specified time intervals. A reasonable way to reach interval statistics is to introduce a time-weighting. The candidate statistics are aggregated over a time interval $[f(d), g(d)]$, where d is again the delivery starting time of the chosen product, and f, g are functions of time with $f < g$. Then, we define the following interval statistics for the product with delivery start d and type i :

- Mid price: The time-weighted mid price $m^{d,i}$ in the interval $[f(d), g(d)]$ is defined as

$$m^{d,i} = \frac{1}{g(d) - f(d)} \int_{[f(d), g(d)]} m_s^{d,i} ds = \sum_{j=0}^J m_{s_j}^{d,i} \frac{(s_{j+1} - s_j)}{g(d) - f(d)}, \quad (4)$$

where $m_s^{d,i}$ is the mid price at time s , i.e. $m_s^{d,i} := \text{bid}_s^{*,d,i} + \frac{\text{ask}_s^{*,d,i} - \text{bid}_s^{*,d,i}}{2}$. As the mid price only changes finitely often in the interval $[f(d), g(d)]$, the integral can be rewritten as a sum. Finally, the ordered set of time points with changes in that interval is denoted by $T_{\text{mid}}^{d,i}$ and contains J elements $s_1, \dots, s_J$. We define $s_0 := g(d)$ and $s_{J+1} := f(d)$.

- Spread: The time-weighted bid-ask spread $\eta^{d,i}$ in the interval $[f(d), g(d)]$ is defined as

$$\eta^{d,i} = \frac{1}{g(d) - f(d)} \int_{[f(d), g(d)]} \eta_s^{d,i} ds = \sum_{j=0}^J \eta_{s_j}^{d,i} \frac{(s_{j+1} - s_j)}{g(d) - f(d)}, \quad (5)$$

where $\eta_s^{d,i}$ is the spread at time s , i.e. $\eta_s^{d,i} := \text{ask}_s^{*,d,i} - \text{bid}_s^{*,d,i}$. As the spread also only changes finitely often in the interval $[f(d), g(d)]$, the integral can again be rewritten as a sum. The ordered set of time points with changes in that interval is denoted by $T_{\text{spread}}^{d,i}$ and contains J elements $s_1, \dots, s_J$. We define $s_0 := g(d)$ and $s_{J+1} := f(d)$.

- Skewness: In order to measure skewness in the order book, we consider volumes v with $v \in \{10, 20, 30, 40, 50, 60\}$ MW. For each v , we compare how much a trader demanding v MW would pay to how much she would get when instead offering v MW on the market. The difference between both is defined as skewness and is supposed to measure the well-balancedness of the LOB. In avoidance of incorporating the spread into this definition, we deduct the best prices of both sides from all respective following price levels. Consequently, we define skewness at one point in time as

$$S^{d,i}(v) = \sum_{j=1}^N v_j^s (p_j^s - p_1^s) \mathbf{1}_{\sum_{k=1}^j v_k < v} + v_{j^*}^s (p_{j^*}^s - p_1^s) - \sum_{j=1}^M v_j^b (p_j^b - p_1^b) \mathbf{1}_{\sum_{k=1}^j v_k < v} - v_{j^*}^b (p_{j^*}^b - p_1^b),$$

where N is the number of price levels on the sell side and M the number of price levels on the buy side. Consequently, v_j^s is the volume available on the sell side at price level j and price p_j^s . Similar roles on the buy side are taken by v_j^b and p_j^b . Finally, p_{j*} and v_{j*} are the price on the last reachable level and the rest volume which is still not covered from the price levels before. Defined like this, skewness incorporates the statistics price levels and LOB depth. Now, we can define

$$SK^{d,i}(v) = \frac{1}{g(d) - f(d)} \int_{[f(d), g(d)]} S_s^{d,i}(v) ds = \sum_{j=0}^J S_s^{d,i}(v) \frac{(s_{j+1} - s_j)}{g(d) - f(d)}, \quad (6)$$

where $S_s^{d,i}(v)$ is the skewness at time s for volume v . Again, the set of time points with changes in that interval can be denoted by $T_{\text{skew}}^{d,i,v}$ and contains J elements $s_1, \dots, s_J$. We define $s_0 := g(d)$ and $s_{J+1} := f(d)$.

Note that $|T_{\text{mid}}^{d,i}|$, $|T_{\text{spread}}^{d,i}|$ and $|T_{\text{skew}}^{d,i,v}|$ need not be equal. Regarding the defined statistics, the time-weighted mid price is assumed to capture the actual value of the product, whereas the bid-ask spread could be seen as an indicator for how much liquidity is available on the market. Skewness parameters try to capture well-balancedness of the market and therefore are an indicator whether the market is a buyer's or seller's market, i.e. a market where purchasers have an advantage over sellers due to an abundance of goods for sale or vice versa [1].

In order to decide for an adequate time interval to aggregate over, we were inspired by [9]: Here, the forecasting results of one of the constructed naive benchmark models for the ID₃ for hourly products could not be significantly outperformed by the best suggested models. This specific benchmark model only relies on volume-weighted transaction data from the quarter hour before the ID₃ interval. Based on this result, we decide to restrict ourselves to LOB statistics from that quarter hour as well. Furthermore, we also include the volume-weighted transaction price as an interval statistic for further analysis, even though the weighting method is a different one in comparison to the other statistics.

Data and model selection The data we use for further analysis are LOB data as well as transaction data from EPEX SPOT for the time from 01.01.2019 to 15.7.2019. This data set is split in training and test set, with the split appearing after three-quarters of the data set. We present the time series for hourly as well as quarter hourly ID₃ values in Figure 1.

We find that in the order book data set belonging to that time period, in rare cases special behaviours appear due to technical reasons on the side of the exchange, like the order book emptying and being filled only 20 minutes later again, and that three hours before delivery. This is far from the norm and thus, we decided to remove periods where the order book is empty on at least one side for the calculation of mid prices and spreads. Affected time intervals are adjusted accordingly. Furthermore, we see spikes in the test set with values far over 200 that have not appeared in the training set. As we assume these spikes to not be well catchable by any estimator knowing only the presented training data, they are removed for further analysis.

To assess which, if any, of the introduced LOB interval statistics have forecasting abilities regarding the ID₃ prices, we check in a first step whether we find a (linear) relationship between the LOB statistics and the ID₃ values as well as between the LOB statistics themselves. This is analyzed through the usage of scatter plots as well as through a correlation analysis. The scatter plots for hourly products are found in Figure 2, and those for quarter hourly products in Figure 3.

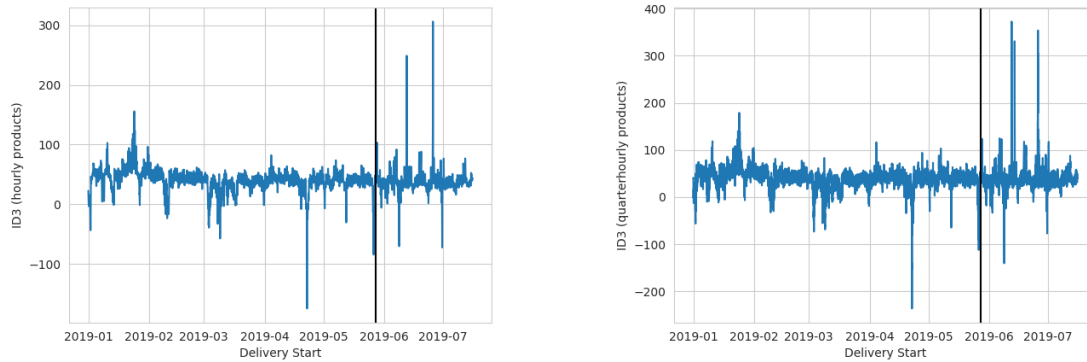


Figure 1: On the left, ID_3 values over a time horizon from 01.01.2019 to 15.07.2019. for hourly products are visible. On the right, ID_3 values for quarter hourly products and the same time horizon are given. In both cases, the black line indicates the split between training set and test set, the former being on the left and the latter on the right of the black line.

Concerning the scatter plots for hourly products, the most prominent feature is the almost perfectly linear relationship between mean transaction prices and mean mid prices. Furthermore, a - less pronounced - linear relationship between both statistics and ID_3 prices is visible. In comparison to that, the mean spread does not seem to have any sort of linear relationship with the ID_3 prices. A possible nonlinear relationship would require other analysis techniques which are beyond the current scope of the paper.

For quarter hourly products, the scatter plots present a similar picture. Mean transaction prices and mean mid prices still show a strong linear relationship, even though it is much less pronounced in comparison to the hourly products. Then again, both have a linear relationship with the ID_3 prices, whereas the spread does not have a visible linear relationship with them. A heatmap is given in Figure 4 for hourly and Figure 5 for quarter hourly products. Again, we notice the correlation of almost 1 for hourly products, and still 0.94 for quarter hourly products when comparing mean mid price and mean transaction price. Furthermore, both statistics show the highest correlation with the ID_3 for hourly as well as quarter hourly products. All asymmetry values show a positive correlation with ID_3 prices, which first grows in numbers with growing asymmetry and then diminishes again. Finally, the spread has a negative correlation with all considered statistics.

Again inspired by [9] and [18], we furthermore conduct an elastic net analysis that was first introduced by [20] for variable and model selection. Elastic nets are a regression method that combines the approaches from the least absolute shrinkage and selection operator (LASSO) and ridge regression. Both are regression techniques that incorporate some sort of regularization, with LASSO also supporting variable selection. LASSO uses an l_1 regularization, ridge regression an l_2 regularization, and elastic nets combine both by summing them in a convex combination with choosable parameter. Furthermore, the decision for the usage of any regularization at all is made by choosing a second parameter α , with very small values for α indicating a negligible influence of the regularization terms. We choose this method due to the strong correlation between transaction prices and mid prices, as ridge regression is known to handle that very well, while LASSO reduces unnecessary variables' coefficients to zero. The needed parameters can both be determined by cross validation techniques that are implemented in the SciKit-Learn Python package presented in [10].

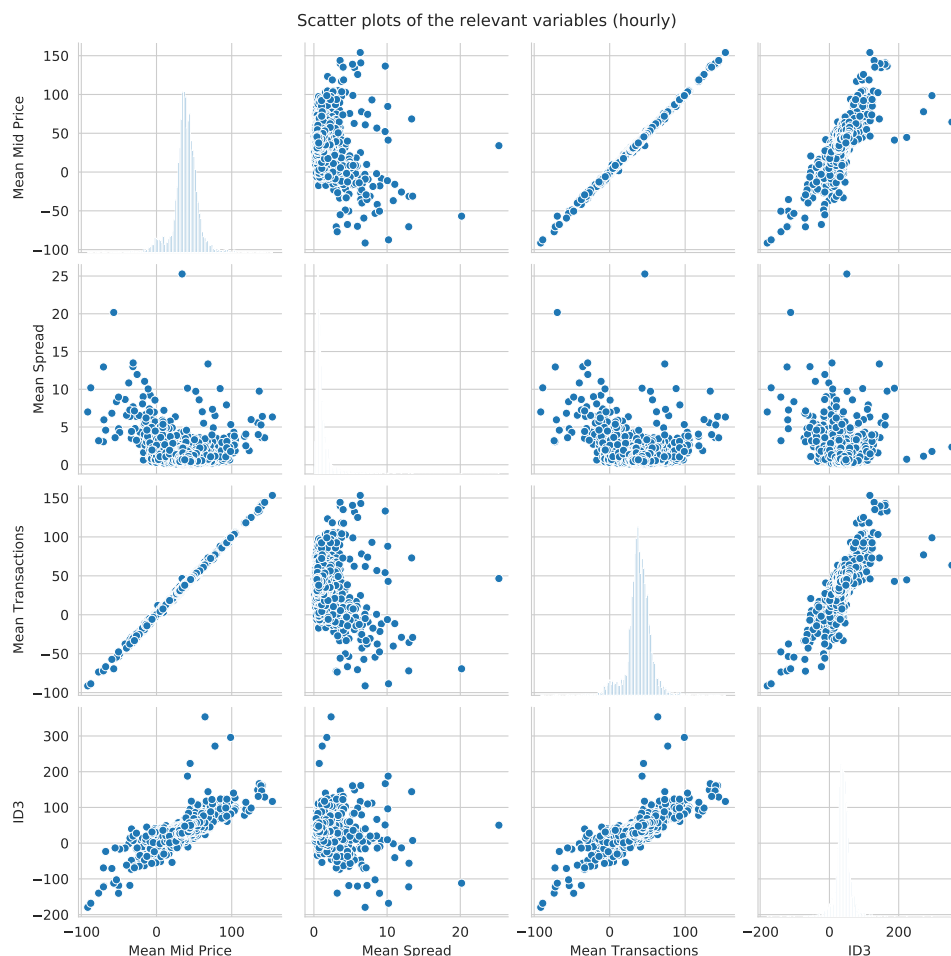


Figure 2: Scatter plots for LOB statistics and the ID₃ for hourly products, excluding order book asymmetry.

We will analyze the chosen coefficients for models trained on different time horizons: First, we look at the model chosen based on the whole training set, then at the models for every month in the training set and finally for every week. This procedure is chosen to capture possible time inhomogenities or dependencies in the data. We present the results for hourly products in Table 1, and the results for quarter hourly products can be found in Table 2.

Our main findings are: For hourly products, transaction prices are deemed the most relevant indicator over the whole time horizon, closely followed by mid prices. This pattern repeats when looking at the monthly and weekly values, with a fluctuation between mostly mid prices and transaction prices regarding the major coefficient. When this is not the case, instead mid prices gain importance. Overall, they are incorporated in the models from time to time, but - most likely due to the strong linear relationship with transaction prices - we do not see a convergence of coefficients here. The spread is also included in roughly half of the models, but the sign of its coefficient varies. Same holds true for all asymmetry values. Therefore, including them in the model does not seem reasonable. Adding to the technical point of view, the models all decided for $\alpha \leq 0.1$, and consequently the resulting elastic net regression is close to an unregularized ordinary least-squares regression.



Figure 3: Scatter plots for the LOB statistics and the ID_3 for quarter hourly products, excluding order book asymmetry.

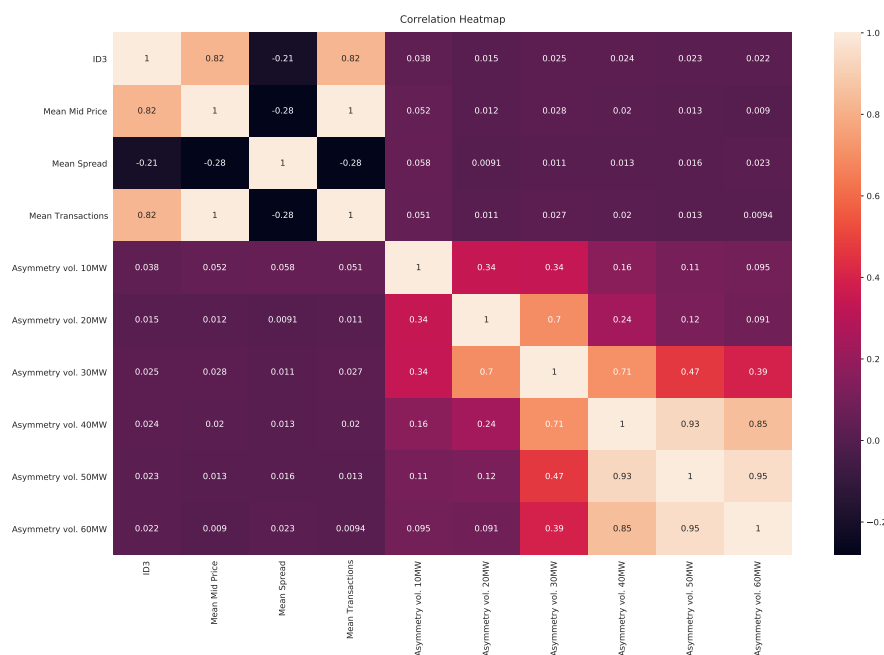


Figure 4: Correlation heatmap for LOB statistics and the ID_3 for hourly products.

Time	Mid Price	Transactions	Spread	As. 10	As. 20	As. 30	As. 40	As. 50	As. 60
none	0.6	0.4	-0	0	0	0	0	0	0
Month 1	0.2	0.77	-0	-0.03	-0	0	0	0.01	0
Month 2	0.11	0.88	-0.06	0	-0	-0	0	0	-0
Month 3	0.78	0.19	0	0.02	0	0.01	0	0	0
Month 4	1.1	0	0	0	0.01	0	0	0	0
Month 5	0.54	0.52	-0.1	-0	-0	-0	-0	-0	-0
Week 1	0	0.95	-0.57	-0.03	0	-0.03	-0	0.02	0
Week 2	0.52	0.5	1.03	-0.27	0.08	0.01	0	0	0
Week 3	0.45	0.44	0	-0	0	0	0	0	0
Week 4	0.28	0.67	0	-0	-0	0	0	0	0
Week 5	0.54	0.49	1.32	0.09	-0.03	-0	-0.02	0	0.01
Week 6	0	0.98	-0	0	0	0	0	0	-0
Week 7	0.84	0	-0	-0	-0	-0	0	0	0
Week 8	0	0.93	0.99	0	0	0	0	0	0
Week 9	1.12	0	-0	0.03	0.02	0	0	0	-0
Week 10	0.42	0.43	-0	0.05	0.01	0.01	0	0	-0
Week 11	0.78	0.12	-0	-0	0	0	0	0	0
Week 12	0.48	0.54	1.76	0.01	-0.06	0.01	0.03	0	-0.01
Week 13	0.88	0	-0	0	0	0	0	0	0
Week 14	1.09	0	1.44	-0	0	0	0	0	0
Week 15	0	0.94	-0	0	0.03	0	-0	-0	-0
Week 16	0.92	0	-0	0	0	0	-0	-0	-0
Week 17	1.05	0.09	0	0	0.03	0	-0	0	0
Week 18	0.66	0.36	-0	-0	-0	-0	0	0	0
Week 19	0.64	0.4	-0.9	0.07	-0.03	-0	0	0	0
Week 20	0	0.89	-0	-0	-0	-0	-0	-0	-0
Week 21	1.16	0	0	0	-0	-0	-0	-0	-0
Week 22	0	0.76	-0.27	0	0	0	0	0	0

Table 1: Values assigned by elastic net to the different variables for hourly ID₃ products. The considered time intervals are the whole training set, each month and each week.

Time	Mid Price	Transactions	Spread	As. 10	As. 20	As. 30	As. 40	As. 50	As. 60
none	0.65	0.28	-0	0	0	0	0	0	0
Month 1	0.46	0.43	-0.06	-0	-0	-0	-0	0	0
Month 2	0.91	0.01	0.22	-0	-0	0	-0	0	-0
Month 3	0.79	0.09	0	-0	0	0	0	0	0
Month 4	0.61	0.46	0.1	-0	0	0	0	0	0
Month 5	0.87	0.06	0	0	0	0	0	0	0
Week 1	0.63	0.25	-0.18	-0	-0	0	0	-0	0
Week 2	0.71	0.28	0.22	-0	-0	-0	-0	-0	-0
Week 3	0.58	0.2	0	0	0	0	0	0	0
Week 4	0.57	0.41	-0.18	-0	-0	-0	-0	-0	0
Week 5	0.69	0.25	0.06	0	-0	-0	-0	-0	-0
Week 6	0.91	0	0.11	-0	-0	0	0	0	0
Week 7	0.52	0.25	0.13	0	-0	0	0	0	-0
Week 8	0.69	0.12	0	0	-0	-0	-0	-0	-0
Week 9	0.95	0.17	-0.09	-0	0	0	0	-0	0
Week 10	0.54	0.26	-0.05	0.01	-0	0	0	-0	0
Week 11	0.84	0	0.06	-0	-0	-0	0	0	0
Week 12	0.87	0.02	0	0	0	0	0	0	0
Week 13	0.62	0.23	-0	-0	0	0	0	0	0
Week 14	1.06	0.06	0	-0	0	0	0	0	0
Week 15	0.5	0.34	0.02	-0	-0	0	0	0	0
Week 16	0.89	0.02	-0.18	-0	0	0	0	0	0
Week 17	0.71	0.53	0.44	-0	-0	-0	-0	-0	0
Week 18	1.05	0	-0	0	-0	0	0	0	-0
Week 19	0.85	0.21	-0.38	0	-0	-0	0	0	0
Week 20	0.7	0.17	-0.15	-0	0	0	0	0	-0
Week 21	0.92	0	0.13	0	0	0	0	0	0
Week 22	0.37	0.3	-0	-0	-0	-0	-0	-0	-0

Table 2: Values assigned by elastic net to the different variables for quarter hourly ID₃ products.
The considered time intervals are the whole training set, each month and each week.

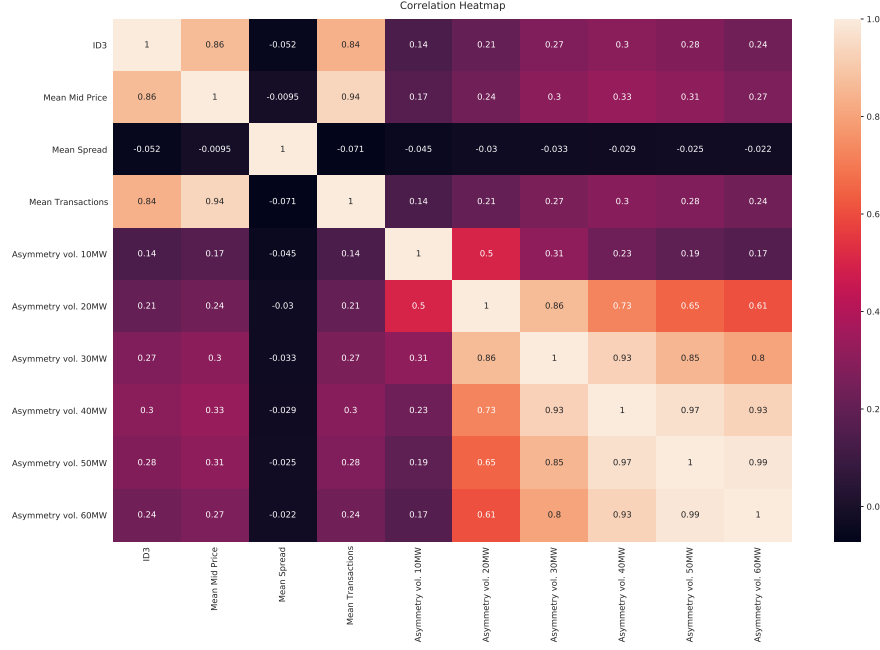


Figure 5: Correlation heatmap for LOB statistics and the ID₃ for quarter hourly products.

For the quarter hourly products, the algorithm again chooses α very close to zero, we find $\alpha \leq 0.05$ for all considered constellations. Therefore, the regression is nearly identical to ordinary-least squares. In comparison to the hourly products, now the main component of nearly all models is the mid price, followed by transaction prices. The spread again is incorporated in more than half of the considered models, but its sign changes and thus, we will not include it in our further analysis. Asymmetry values now are almost all equal to zero and are also not considered any further.

Based on this analysis, we decide to drop all explanatory variables except mid prices and transaction prices for both hourly and quarter hourly products. Furthermore, we will not combine these two variables in one model due to the strong correlation between them, but instead compare them against each other in their predictive ability.

The two naive models Based on the previous analysis, we present in a first step two naive models, where one is based on mid prices and the other is based on transaction prices. The former is defined by

$$\text{mid ID}_3^{d,i} = \sum_{s_j \in T_{\text{mid}}^{d,i} \cup g(d)} m_{s_j} \frac{(s_{j+1} - s_j)}{g(d) - f(d)}, \quad (7)$$

where all variables are defined as in (4). The second model that is based on transaction prices is denoted as

$$\text{trans ID}_3^{d,i} = \sum_{t \in T_{\text{trans}}^{d,i}} \frac{p_t v_t}{\sum_{t \in T_{\text{trans}}^{d,i}} v_t}, \quad (8)$$

where p_t is the price of transaction t and v_t is the corresponding traded volume. $T_{\text{trans}}^{d,i}$ contains all transactions that take place in the interval $[f(d), g(d)]$ for products with type i and delivery start d . We again define $d = (T, H)$ as time stamp containing date T and hours H of the delivery start. Furthermore and in correspondence with the definitions before, i is again contained in $\{h, qh\}$, representing either hourly or quarter hourly products. In case of no transactions existing

in the relevant quarter hour, the forecasted value is replaced by the corresponding day-ahead price.

Now, what is left is to define f and g to determine start and end time of the interval. As we want to use the quarter hour before the start of the ID_3 interval, we define

$$f(d) := (T - s, (H - 3.75) \bmod 24)$$

and

$$g(d) := (T - s, (H - 3.5) \bmod 24).$$

Here,

$$s = \begin{cases} 1 & (H - 3.5) \leq 0, \\ 0 & \text{else.} \end{cases}$$

Again with the example of the ID_3 of the hourly product 8 am - 9 am, the transactions from 4:15 am to 4:30 am are contained in $T_{\text{trans}}^{d,i}$. With this definition, ${}_{\text{trans}}\hat{ID}_3^{d,i}$ equals the naive benchmark model from [9] mentioned above.

It is to mention that, even though ${}_{\text{trans}}\hat{ID}_3^{d,i}$ performs very well on hourly products as seen in [9], its performance regarding quarter hourly products fell behind the results of the other presented models in the paper. Nonetheless, it serves well as a benchmark to assess whether LOB statistics can contribute to ID_3 forecasting.

Naive forecast We conduct a first analysis by using the models ${}_{\text{mid}}\hat{ID}_3^{d,i}$ and ${}_{\text{trans}}\hat{ID}_3^{d,i}$. The error of the forecasted ID_3 prices in comparison to the actual observed ID_3 prices is calculated via mean absolute error (MAE) as well as via root mean squared error (RMSE), which are defined by

$$MAE(\hat{P}^i) = \frac{1}{|D^i|} \sum_{d \in D^i} |ID_3^{d,i} - \hat{P}^{d,i}|, \quad (9)$$

$$RMSE(\hat{P}^i) = \sqrt{\frac{1}{|D^i|} \sum_{d \in D^i} (ID_3^{d,i} - \hat{P}^{d,i})^2}, \quad (10)$$

where D^i is the set of all timestamps in the test set that a forecast is calculated for in regard to product type i , and $\hat{P}^{d,i} \in \{{}_{\text{mid}}\hat{ID}_3^{d,i}, {}_{\text{trans}}\hat{ID}_3^{d,i}\}$. Both are regularly used as error measures in the literature, see e.g. [9] or [4]. These error measures indicate differences in forecasting accuracy between both estimators. We present them for both models in Table 3.

Naive Model	MAE _h	RMSE _h	MAE _{qh}	RMSE _{qh}
${}_{\text{mid}}\hat{ID}_3^{d,i}$	4.02	10.64	6.04	12.52
${}_{\text{trans}}\hat{ID}_3^{d,i}$	4.01	10.62	6.72	13.78

Table 3: Forecasting errors for both considered naive models.

The results show that in the naive version of the forecasts, ${}_{\text{mid}}\hat{ID}_3^{d,h}$ and ${}_{\text{trans}}\hat{ID}_3^{d,h}$ perform similarly well. This is supported by an analysis with the adjusted Diebold Mariano test from [2], where we cannot reject the hypothesis that both forecasts have the same forecasting accuracy. In contrast to that, ${}_{\text{mid}}\hat{ID}_3^{d,qh}$ and ${}_{\text{trans}}\hat{ID}_3^{d,qh}$ show a bigger gap in performance, with mid prices outperforming transaction prices. Again, though, that difference in forecasting accuracy is not deemed statistically significant.

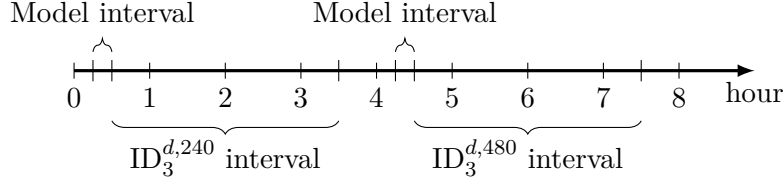


Figure 6: Example of available calibration data while forecasting the ID₃ for the hourly product 8 am - 9 am. It is visible that at the beginning of the model interval, only information from four or more hours earlier is available.



Figure 7: Example for rolling calibration window with window size $w = 4$. The blue dots represent the calibration set, the green one is the point where the forecast is taking place and the calibration is used, and the gray dots are data points that are not in usage in that iteration.

Regression forecast A simple way to increase the complexity of both models is to add an intercept as well as a slope to the models, i.e.

$$\tilde{P}^{d,i} = a_0^i + a_1^i \hat{P}^{d,i} \quad (11)$$

for $\hat{P}^{d,i} \in \{\text{mid}\widehat{\text{ID}}_3^{d,i}, \text{trans}\widehat{\text{ID}}_3^{d,i}\}$ and a_0^i, a_1^i the regression parameters for product type i . For the resulting new models, the first step is to find adequate sizes for the calibration windows of the regression parameters. [7] present a model for the hourly ID₃ and test which calibration window size fits the model best. Their findings indicate that for ID₃ prices, the longest possible calibration window indeed performed the best with their forecasting model. To choose a calibration window size for our new models, we perform a rolling window approach on the whole training data set, repeated with differing window sizes $w \in [1, 3000]$. In order to let all windows roll over the same data set, we start after the first 3000 data points, which are then used for the first calibration. This approach is detailed in the following: At every step, we forecast one point in time only. For each new hourly ID₃ price, new information is available that is included in the calibration. It is important to notice, though, that the most recent information tuple available for calibration are the ID₃ price and its estimate from four hours before, see Figure 6.

Now, for each calibration window size w , the calibration set as well as when the forecast takes place is rolled through the entire data set. This is depicted in Figure 7. The resulting mean absolute errors for $w \in [1, 3000]$ on the training data set and models $\text{mid}\widehat{\text{ID}}_3^{d,h}, \text{trans}\widehat{\text{ID}}_3^{d,h}$ are given in Figure 8 for hourly and in Figure 9 for quarter hourly products. For hourly products, they indicate the same behaviour as described by [7], i.e. the bigger the calibration window, the better the forecast. In contrast to that, though, we also notice small errors for smaller window sizes around 260. These two patterns lead us to the decision to choose calibration window sizes of $w \in \{260, 3000\}$ for comparing the models on the test set for hourly products. The decision against even bigger calibration windows is due to the total size of the training data set, as increasing the calibration window is equivalent to decreasing the number of points for forecasting in the training set.

For quarter hourly products, the picture is similar when looking at big calibration window sizes for $\text{trans}\widehat{\text{ID}}_3^{d,qh}$, where we find the lowest error values. Interesting enough though, we see a strong dip in the MAE for a window size of $w = 3$ for $\text{mid}\widehat{\text{ID}}_3^{d,qh}$. Consequently, we decide for

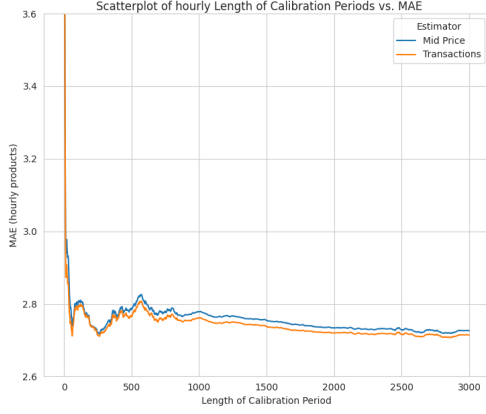


Figure 8: Mean absolute errors for both estimators on the training set for growing calibration window size and hourly products.

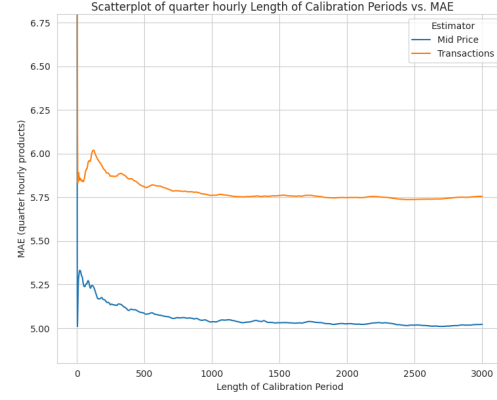


Figure 9: Mean absolute errors for both models on the training set for growing calibration window size and quarter hourly products.

calibration window sizes of $w \in \{3, 3000\}$ for the test set of quarter hourly products.

Based on the calibration windows found in this first step, we follow the steps from before and perform a forecast analysis. We again use MAE as well as RMSE for forecasting evaluation on the test set, with w defined as above. Results are given in Table 4 for hourly and in Table 5 for quarter hourly estimators.

Regression Model	MAE _h ($w = 260$)	RMSE _h ($w = 260$)	MAE _h ($w = 3000$)	RMSE _h ($w = 3000$)
$\widetilde{\text{midID}}_3^{d,i}$	4.32	10.71	4.06	10.63
$\widetilde{\text{transID}}_3^{d,i}$	4.31	10.69	4.05	10.62

Table 4: Forecasting errors for all hourly regression models.

Regression Model	MAE _{qh} ($w = 3$)	RMSE _{qh} ($w = 3$)	MAE _{qh} ($w = 3000$)	RMSE _{qh} ($w = 3000$)
$\widetilde{\text{midID}}_3^{d,i}$	6.38	14.04	6.13	12.56
$\widetilde{\text{transID}}_3^{d,i}$	7.05	15.03	6.81	13.82

Table 5: Forecasting errors for all quarter hourly regression models.

We find a pattern similar to the one for the naive forecast for hourly products, with the transaction price model outperforming the mid price model. When comparing the values to the errors reached by the naive models in Table 3, we see no improvement regarding MAE. For the RMSE, though, both regression approaches outperform or at least equal its naive counterpart for $w = 260$.

For quarter hourly products, our findings mirror the results of the naive approach: Again, the mid price model outperforms the transaction price model. Nonetheless, it holds true for both estimators that they are dominated by their respective naive counterpart regarding both MAE

Model	MAE _h pos.	RMSE _h pos.	MAE _h neg.	RMSE _h neg.	MAE _h reg.	RMSE _h reg.
$\widehat{\text{ID}}_3^{d,i}$	86.95	111.94	11.19	12.59	3.39	5.13
$\widehat{\text{ID}}_3^{d,i}$	85.87	111.57	11.32	12.45	3.38	5.16
$\widetilde{\text{ID}}_3^{d,i}$ with $w = 3000$	86.56	111.64	11.09	12.38	3.43	5.16
$\widetilde{\text{ID}}_3^{d,i}$ with $w = 3000$	85.49	111.27	11.23	12.25	3.42	5.18
$\widetilde{\text{ID}}_3^{d,i}$ with $w = 260$	85.97	111.31	10.74	11.75	3.69	5.38
$\widetilde{\text{ID}}_3^{d,i}$ with $w = 260$	84.85	110.9	10.84	11.53	3.69	5.4

Table 6: Forecasting errors for all models and three price regimes for hourly products.

Model	MAE _{qh} pos.	RMSE _{qh} pos.	MAE _{qh} neg.	RMSE _{qh} neg.	MAE _{qh} reg.	RMSE _{qh} reg.
$\widehat{\text{ID}}_3^{d,i}$	75.3	95.88	21.64	25.6	5.24	7.92
$\widehat{\text{ID}}_3^{d,i}$	82.42	100.42	23.77	28.93	5.84	9.28
$\widetilde{\text{ID}}_3^{d,i}$ with $w = 3000$	74.93	95.6	21.68	25.7	5.32	8.01
$\widetilde{\text{ID}}_3^{d,i}$ with $w = 3000$	82.32	100.38	23.52	28.73	5.93	9.35
$\widetilde{\text{ID}}_3^{d,i}$ with $w = 3$	57.35	76.07	29.25	33.5	5.71	11.6
$\widetilde{\text{ID}}_3^{d,i}$ with $w = 3$	62.8	78.51	30.45	36.75	6.32	12.58

Table 7: Forecasting errors for all models and three price regimes for quarter hourly products.

and RMSE.

In order to dig deeper into these observations and inspired by the lower RMSE errors for hourly products for the regression estimators, we repeat the error measurements for different price regimes in order to analyze which of the models performs best in different market situations.

Forecast errors for price regimes Electricity index prices, as seen in Figure 1, tend to be stable around a certain value, with sometimes abrupt jumps up- or downwards. In order to analyse which model performs best in these differently behaved market situations, the first step is to split the observations into defined regimes. Here, we follow [18] and introduce three price regimes: Based on the mean and three standard deviations rule, we define positive, negative, and regular price regimes through

$$\text{pos.}: \mu^i + 3\sigma^i < \text{ID}_3^{d,i}, \quad \text{neg.}: \mu^i - 3\sigma^i > \text{ID}_3^{d,i}, \quad \text{reg.}: \mu^i + 3\sigma^i \geq \text{ID}_3^{d,i} \geq \mu^i - 3\sigma^i.$$

The data in the test set is split into these groups and errors are calculated separately for each of them. Our analysis includes the naive models as well as the regression models. We present results for hourly products in Table 6 and for quarter hourly products in Table 7.

We find that for hourly products and the regular price regime, the naive models perform best as was to be expected based on all former results. For all error measures regarding positive or negative price regimes though, the regression models outperform their naive counterparts. Here, we notice that in the case of the negative price regime and MAE, the mid price model actually outperforms the transaction price model. Nonetheless, this outperformance is not of significant size.

Furthermore, we find that for quarter hourly products and positive price regimes, the regression models again outperform their naive counterparts, which is most visible for both estimators with $w = 3$. In comparison to the hourly products, though, we don't see an improvement for the negative price regime. The regular price regime finally is again dominated by the naive estimators. These findings indicate that in market situations based on the positive price regime, we see a stronger autocorrelation in the prices in comparison to negative or regular price regimes.

5. Conclusion

We have considered the question whether the usage of limit order book statistics has the potential to enhance existing models for ID₃ prices. Based on our analysis, our answer is that LOB derived time-weighted mid prices indeed contain much information about index prices. For hourly products, this information does not seem to enhance the forecasting performance notably, as our model is slightly outperformed by the naive benchmark. Furthermore, as working with LOB data requires more computational resources than working with transaction data only, we do not suggest this course for hourly products. In comparison though, the time-weighted mid prices seem to carry more information than the transaction prices for quarter hourly products, which can be seen by an outperformance of the naive benchmark in regard of MAE and RMSE. A simple increase of model complexity through usage of a regression did not yield the anticipated improvement of forecasting accuracy. Nonetheless, we find that for extreme price regimes, regression models tend to outperform their naive counterparts, which was strongly visible with quarter hourly products and the positive price regime. Consequently, adding information from derived LOB statistics - especially time-weighted mid prices - to existing full information models is the obvious next step in order to enhance other electricity market intraday indices. Furthermore, our results show that a strongly structural different model - relying on a time-weighting and neglecting every volume-related information - can contain more information than the volume-weighted transaction data themselves, which in itself might be worth of further research.

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A. EPEX SPOT Order Book Data

This section introduces the structure of an order a trader places on the intraday market. Furthermore, we present the structure of order book data for EPEX SPOT data, and it is explained in detail which information is available during its usage. Finally, the subtleties and pitfalls of the data set are discussed. As limit orders are mostly used on the market, we concentrate on them in the following section and ignore market sweep orders.

At the intraday market operated by EPEX SPOT, a trading system called M7 is used. Aside from entering a side (buy or sell), the product of interest as well as a price and a quantity, it enables the traders to make several specifications for limit orders. Orders can be specified regarding type, execution restriction and validity restriction. In the following, we present the offered alternatives for these three specification types. A reader familiar with the structure of orders on the intraday market may well skip this first passage.

The exchange offers different types of orders that a trader chooses from:

1. Regular Order (REG): Regular orders are regular limit orders that are placed with a certain limit price and a quantity. They have to be executed at their specified limit price or better.
2. Iceberg Order (ICB): An iceberg order is a certain type of limit order which is used to conceal a bigger trade in order to limit the market movement against the trading party. Only a part of the total wanted quantity is visible to the market participants, even though the full quantity is available to them. An order with beforehand-specified quantity is placed at the market, and after matching an existing or incoming order, a next order with the same quantity is placed. This goes on until deletion of the iceberg order or complete fulfillment.
3. Indicative Order (IND): This order provides an indication that the trader is willing to negotiate around the specified price for the chosen delivery instrument.
4. Over-the-counter order (OTC): This is an order with a predesignated counterpart, i.e. a trade which is not available to all market participants.

Furthermore, an execution restriction can be added to the order:

1. All or None (AON): The order is either matched in its entirety or not matched at all. This order type is not found in the regular order book, but solely in the order book for user-defined block orders.
2. Fill or Kill (FOK): The order is either immediately matched in its entirety or immediately deleted after submission.
3. Immediate or Cancel (IOC): The order must be immediately executed or is deleted. In comparison "fill or kill", it is possible to only partially match other orders.

Finally, the trader has the possibility to choose a validity restriction for the order:

1. Good for Session (GFS): The order is deleted when the trading end time of the contract is reached.
2. Good till Date (GTD): The order comes with an additional information containing the time and date at which to delete the order.

Another very important choice for the trader is in which book she wants to place her order: In the book for local orders or in the book for remote orders. Local orders are matched locally in M7, whereas remote orders are transferred to the XBID System which has exclusive access to cross-border capacities. In the latter case, orders can be matched with orders in another country if cross-border capacity allows it. As the working hours for the XBID market are a subset of the M7 working hours, traders can first enter their orders locally, then remove them and insert them remotely, and finally remove them again and insert them locally again.

All the listed specifications lead to a basket of information that is attached to every order. Some of this information is contained in the data sets for order book data available from EPEX SPOT. Of course though, their structure has seen changes over the last years, as the intraday market itself has. Therefore, we explain in detail what information the latest data format D2 (applied from November 2019 onward) as well as its predecessor D1 (encompassing the period from 2017 to November 2019) contain. The elements contained for both types are listed in Table 8 below. Columns corresponding to each other are placed in the same row. Some columns don't have an exact equivalent, but their information is nonetheless found in the other data type as well, e.g. Delivery Start and Delivery End for D2 can be translated to Delivery Date and Delivery Instrument and Instrument Type of D1. In comparison, some columns of D2 contain information that is not found in D1, namely Action Code, Currency, isOTC, RevisionNo and Execution Restriction.

An innovation presented in D2 is the introduction of action codes: In the old data format, a row contained always at least two different events for one order ID, i.e. activation and deactivation of the order, possibly also a timestamp for a modifying event. The new format has a different approach: Every row accounts for one event only, and the type of event is denoted by an action code. The action code specifies which action is applied to the order. The different possible actions and their typical appearances are listed in the following:

- A for Activation : Following an order through its lifetime, its first appearance always starts with its activation. An order with action code A also appears after deletion of the previous order and creation of a (slightly) changed one.
- X for Automatic Deletion: This action determines the end of the lifetime of an order and is performed by the system itself. Nonetheless, the order could still be parent order of a new order appearing afterwards.
- D for Manual Deletion: This action also determines the end of the lifetime of an order, with the small difference that now the order was manually deleted. Again, the order could still be parent order of a new order appearing afterwards.
- M for Matched: If an order is completely matched, its quantity is denoted by 0.0 and its action code is M. An M furthermore determines the end of the lifetime of an order.
- P for Partially Matched: In the case of a partial matching of the order, the corresponding action code is P and the quantity associated with it is the remaining quantity after the partial match. If the corresponding order is an Iceberg Order, even a complete fulfilment of the stated quantity is commented with action code P. This goes on until complete fulfilment (M) of the Iceberg Order or until its deletion (X or D).

Columns D2	Columns D1
Order ID	Order ID
Initial ID	Initial ID
Parent ID	Parent ID
Entry Time	Start Validity Date
Transaction Time	End Validity Date
Validity Time	Cancelling Date
Action Code	
Delivery Start	
Delivery End	
	Delivery Date
	Delivery Instrument
	Instrument Type
Product	Area
Delivery Area	Area
Market Area	Side
Side	Price
Price	
Currency	
Quantity	Volume
isOTC	
RevisionNo	
Is User Defined Block	Is Block
Execution Restriction	
	Is Executed
	Execution Price
	Execution Volume

Table 8: Columns contained in the data formats D2 and D1. Columns found in both data formats are contained in the same row, those without an equivalent stand alone.

- I for Insert: If one shown share of an Iceberg Order is matched (with a P), the next equally sized limit order is placed at the market through an insertion I. In this case, the order ID stays the same. This procedure goes on until complete fulfilment (M) of the Iceberg Order or until its deletion (X or D).
- C for Changed: This represents technical changes in the M7 trading system and does not account for any real modification. This leads to the counter-intuitive behaviour that a modification of the order limit is not denoted with C, but instead the order is deleted and a new order gets activated.
- H for Hibernated: Hibernation temporarily deactivates an order until further notice.

The life story of one order is thus presented by filtering for the order ID of interest and then sorting after revision number.

Listing the advantages of the new data format, one might assume it to be the better fit for building LOBs or similar tasks. Nonetheless, one topic impedes this: Each order comes with several timestamps, namely an entry time, a transaction time and sometimes a validity time. The entry time is the time used in the price/time ranking in M7, and should therefore also be used for building LOBs. This timestamp, however, struggled with inconsistencies in the export from M7 to the database especially for historical files¹. That behaviour is supposed to change in one of the upcoming M7 releases, but up to now this prevents from using the new data format to reliably build LOBs.

The orders in the old format can be subject to the different order type choices as well as execution and validity restrictions as described above. [8] describe the existence of an execution restriction as a cause of trouble while building LOBs with the old data format from the EPEX SPOT, as the chosen restrictions are not communicated. Nonetheless, we believe this to be easily handled: As AON orders are part of a separate order book, they are not part of the data. FOC and IOC orders, if not filled or partly executed, respectively, have a validity duration of zero and therefore do not enter the LOB. Furthermore, the problem of erroneous order information in columns "execution price" and "execution volume" mentioned in [8] was removed in 2017.

¹This was stated by EPEX SPOT in a mail correspondence from November 2021.